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Application Note

RIsearch and siOFF: An integrated, high-performance framework for RNA–RNA interaction and siRNA off-target assessment

Stefano Roncelli [1], Lorenzo Favaro [1], Christian Anthon [1] and Jan Gorodkin [1,*]

[1]Center for non-coding RNA in Technology and Health, Section for Health Data Science and AI, Department of Public Health, University of Copenhagen, Thorvaldsensvej 57, 1871, Frederiksberg, Denmark

*Corresponding author. gorodkin@rth.dk

Abstract

Motivation: A substantial part of RNA function comes from its interactions with other RNAs. Given the sheer scale of the search space (genome/transcriptome), there is high demand for fast RNA interaction search software that should also be user-friendly and easy to integrate with other tools.

Results: Here, we introduce new versions of RIsearch and its downstream siRNA off-target assessment pipeline. Both tools now have substantially improved usability, interoperability, and installation procedures. RIsearch (v3) is a Rust reimplementation of the RIsearch2 search algorithm, verified against the reference; it reworks the original command-line interface, adds Python bindings, and improves performance. siOFF modernises the previous siRNA off-target pipeline through a redesigned CLI, improved distribution and faster execution.

Availability and implementation: RIsearch and siOFF are released under the Business Source License 1.1 at github.com/RTH-tools/risearch and github.com/RTH-tools/siOFF. Both software packages are available on PyPI as `risearch` and `sioff`; the Rust crate is also indexed on Crates.io as `risearch`.

Contact: Jan Gorodkin, gorodkin@rth.dk

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Introduction and motivation

Intermolecular RNA interactions with both RNA and DNA underpin a wide range of regulatory mechanisms, and demand is increasing for predicting them at genome- or transcriptome-wide scale. While RNA–RNA interactions continue to receive increasing attention, recent work has revealed enormous potential for RNA–DNA interactions (Lambolez et al., 2026). When studying and using endogenous and synthetic small interfering RNAs (siRNAs), the lack of the off-target landscape can confound RNAi knockdown experiments and, for example, limit therapeutic application through unintended interactions with off-targets.

RIsearch2 (Alkan et al., 2017) is a tool designed for large-scale screens of near-complementary RNA–RNA interactions. Because it ignores intramolecular interactions, it is particularly suitable for probing the off-target landscape of small RNAs, like siRNAs. RIsearch combines a suffix-array index of the target with a seed-and-extend dynamic-programming step that uses a nearest-neighbour simplified energy model (Wenzel et al., 2012). The companion siRNA off-target pipeline turns these predictions into per-transcript binding probabilities and a transcriptome-wide off-targeting potential through a partition-function model that integrates transcript abundance and binding-site accessibility.

With the development of modern software ecosystems, new opportunities have emerged to improve how RIsearch2 and its pipeline were originally distributed and used. Here, we introduce new versions of RIsearch2 and the siRNA pipeline, completely redesigned for improved usability, interoperability, installation, and development. RIsearch (v3) is a faster, memory-safe reimplementation of the same core search algorithm, with a redesigned command-line interface and PyO3 Python bindings. The siRNA Off-Target pipeline, siOFF, is a rewrite of the previous siRNA off-target pipeline and can now use the RIsearch (v3) bindings directly; it also serves as an example of using RIsearch embeddings and integrating them into a Python workflow.

A limitation of the original RIsearch2 package is its distribution and installation: it is provided as C source code that each user must compile locally. RIsearch (v3) improves on these aspects while meeting three essential requirements at once:

1. retaining the genome-scale throughput that keeps whole-transcriptome screens practical;
2. installing from a standard package registry, with no manual build;

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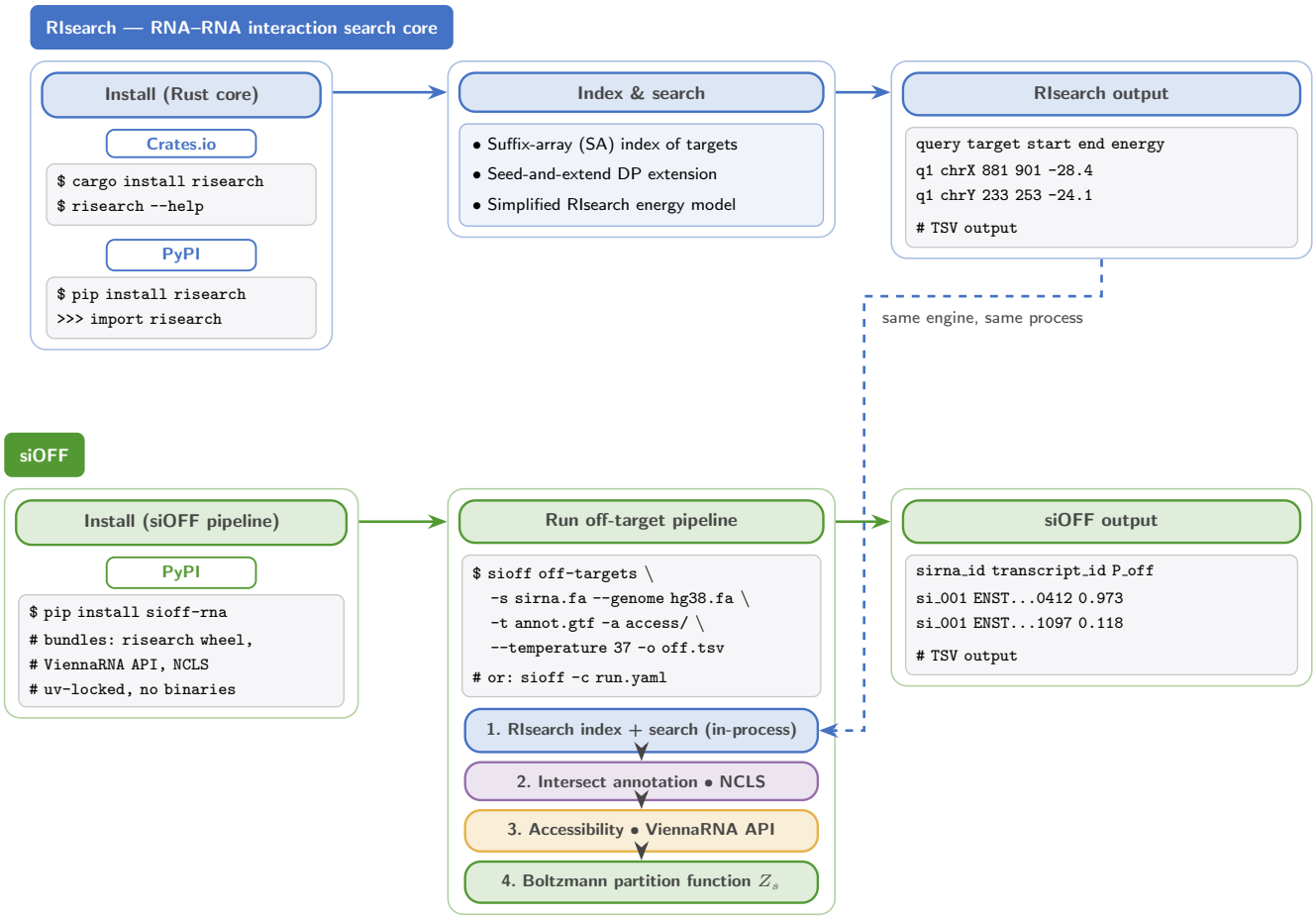


Figure 1 Installation and execution of RIsearch and the siOFF pipeline. (Top, blue) RIsearch is a Rust reimplementation of the RIsearch2 search algorithm, installable via `cargo install risearch` (Crates.io) or as a Python wheel via `pip install risearch` (PyPI). The core builds a suffix-array index of target sequences and runs a seed-and-extend dynamic-programming search using the simplified RIsearch energy model, producing interaction hits as a TSV file. **(Bottom, green) siOFF** (siRNA Off-Target pipeline) is a timely Python 3 rewrite installable with a single `pip install sioff-rna`, which bundles the RIsearch dependencies, the ViennaRNA 2.7.2 Python API, and an in-memory nested containment list (NCLS) interval index with no external binaries required. siOFF orchestrates four stages: (*i*) RIsearch index and search invoked natively via PyO3 bindings; (*ii*) intersection of predicted binding sites with a transcriptome annotation using an NCLS index; (*iii*) per-site RNA accessibility computed through the ViennaRNA Python API; and (*iv*) a per-siRNA Boltzmann partition function yielding off-target probabilities, written to a TSV output file. The dashed arrow indicates that siOFF calls RIsearch within the same process via the Python bindings, with no subprocess or intermediate files.

3. providing a Python library, so that any analysis can depend on it directly.

C reaches that throughput but has no standard package registry of its own, so prebuilt distribution depends on external packaging, and it cannot guarantee memory safety at a library boundary. Python distributes and embeds easily, but pure Python runs the seed-search and dynamic-programming loops in an interpreter, which is far slower than compiled native code and cannot reach that throughput. Rust, however, meets all three: it compiles to fast native code, installs from standard package registries, and can expose the algorithm through a memory-safe Python library. These capabilities are why we rebuilt RIsearch in Rust.

RIsearch reimplementation

RIsearch (v3) keeps RIsearch2's search algorithm unchanged: targets are concatenated and indexed in a suffix array, then each query is scanned against the index for near-complementary seeds. The seeds are then extended in both directions with Gotoh's three-state affine-gap recurrence (Gotoh, 1982), scoring stacked pairs from a nearest-neighbour dinucleotide model that penalises bulges and internal loops. By contrast, the implementation was rewritten from scratch in Rust. Most of the code is safe Rust, which guarantees memory safety; unchecked memory access is confined to the performance-critical inner loops of seed search and extension and to the zero-copy loading of the on-disk index. The suffix array is now built with libsais (Grebnov, 2021), a linear-time construction that can run in parallel across cores, replacing the single-threaded libdivsufsort (Mori, 2015) used by RIsearch2. We regenerated the energy parameter tables from ViennaRNA duplex energies, so the calculated hybridisation energies differ from those of RIsearch2.

Seamless installation

RIsearch2 is distributed as C source code that each user must compile by hand, which requires a C toolchain and leaves build settings and

dependency versions to each machine. The new version needs no manual build: it installs with a single command from the Python and Rust package registries, PyPI and Crates.io, or as a prebuilt binary from each GitHub release.

Improved usability

While RIsearch2 can return many overlapping alignments for one duplex, RIsearch (v3) by default reports one alignment per duplex: seed extensions that end on the same query and target coordinates are collapsed to the one with the lowest (most favourable) energy. For example, two seeds at neighbouring positions of an siRNA that extend to the same duplex yield one row instead of two near-identical ones, unless collapsing is disabled. RIsearch (v3) also validates its inputs more strictly than RIsearch2, rejecting malformed input with an error that names the offending record or line. The command line is rebuilt around task-oriented subcommands, while legacy RIsearch2 flags are still accepted to ease migration of existing scripts.

Enhanced interoperability

RIsearch2 was designed as a standalone command-line tool, so embedding it in a Python analysis meant spawning a subprocess and parsing its text output. The new version of RIsearch instead ships PyO3 bindings in the same wheel: `risearch.index()` builds an on-disk index, `risearch.TargetRegistry.open()` loads it as a reusable memory-mapped handle, and `risearch.search()` returns hits as a Polars DataFrame through the Arrow C stream interface, with no intermediate files and no text parsing:

```
import risearch
risearch.index("targets.fa", "index/")
store = risearch.TargetRegistry.open("index/")
df = risearch.search(queries, store, matrix="t04")
```

Searches run without blocking other Python threads in the same process, and the energy parameter set can be chosen per call.

Extensible energy parameters and engine

The energy parameters are stored as tables separate from the search code: the bundled sets are generated from ViennaRNA by a script in the RIsearch repository, and users can load their own table at runtime, so new thermodynamic parameters require no change to the search engine and no knowledge of Rust internals. The engine itself is modular, and its behaviour is verified against independent reference implementations through property-based and metamorphic tests, complemented by mutation testing and bounded model-checking proofs (Kani), so contributors can extend it with confidence.

Faster execution time

On an AMD EPYC 9554 server, with both tools searching the same 63 siRNAs against the same chromosome-X transcript set, RIsearch (v3) is 2–3× faster than RIsearch2 (Supplementary Figures S1–S4; Supplementary Results S1.1). The suffix array creation rewrite also yielded about 2× faster execution times (Supplementary Figure S5; Supplementary Results S1.2). Because the on-disk index is now memory-mapped and read without deserialisation, multi-gigabyte genome indexes are available immediately and shared across processes by the operating system, rather than read into memory on every run.

siOFF: A reimplemented siRNA Off-Target assessment pipeline

Alongside the new version of RIsearch, we distribute **siOFF**, a complete reimplementation of the siRNA off-target pipeline introduced by Alkan et al. (2017) (Fig. 1). Beyond porting the workflow from the end-of-life Python 2.7 to Python 3, the rewrite dismantles the concrete obstacles that limited adoption of the original pipeline: an installation that required a hand-compiled RIsearch2 binary and two external executables, a rigid single-script interface with a fixed set of flags, and subprocess-and-file coupling that prevented embedding in larger analyses. The underlying four-stage thermodynamic model—search, annotation intersection, accessibility, and a per-siRNA Boltzmann partition function—is preserved unchanged and described in full in S2.1.

One-command installation

The original pipeline requires a Python 2.7 interpreter, a separately compiled RIsearch2 C binary, and installation of external software `RNAplfold` (ViennaRNA) and `bedtools` (Quinlan and Hall, 2010). siOFF removes the need for external dependency installation: a single `pip install sioff-rna` brings in the RIsearch software as a pre-compiled wheel, the ViennaRNA 2.7.2 Python API (Lorenz et al., 2011) in place of the `RNAplfold` executable, and an in-memory nested containment list (NCLS) (Alekseyenko and Lee, 2007)—an interval index that answers genomic overlap queries directly in Python—in place of `bedtools`. All transitive dependencies are resolved and pinned through a universal `uv` lockfile, so installations are reproducible and free of version conflicts across machines.

Easier to use

The legacy pipeline was a single `argparse` script driven by a fixed set of command-line flags. siOFF retains backwards-compatible flags but adds a YAML configuration mode: every option—inputs, the RIsearch energy model, seed geometry, and energy threshold, the saturation parameters α, γ, and θ, and the output format—can be declared in one file, validated at startup against a typed OmegaConf schema. A single configuration file captures an entire run, making analyses straightforward to reproduce and to share between collaborators. For large screens, siOFF integrates natively with Slurm: one orchestrator submits the indexing, accessibility, and off-target stages as dependent batch jobs with per-stage resource requests, and a single configuration can fan one screen out across multiple transcriptomes as independent job groups, replacing hand-written submission scripts.

We have now implemented a single temperature option (default 37 °C). The temperature propagates to both the ViennaRNA accessibility calculation and the Boltzmann partition function, so that hybridisation and opening energies are always evaluated on a consistent thermodynamic scale—directly replicating the intent of the original RIsearch model (Wenzel et al., 2012) while exposing the folding condition as a first-class parameter (Supplementary Methods S3).

Easy integration

Because the legacy pipeline communicated through intermediate files and subprocesses, embedding it in a larger analysis meant shelling out to a script and parsing its text output. siOFF instead exposes its stages as an importable Python API: the search, intersection,

accessibility, and probability services can each be called directly and return native Polars DataFrames, so a user can drop any stage into an existing Python workflow and pass results to other Python-native tools in a single process, with no intermediate files and no text parsing. Based on RIsearch software bindings (Section 2), siOFF thus serves both as a turnkey command-line tool and as a library component of off-target ad hoc analyses.

Enhanced execution time

The pipeline was benchmarked on an AMD EPYC 9554 (64 cores, 32 workers) using subsets of the genome-wide siRNA efficacy library of Huesken et al. (2005) searched against the GRCh38/hg38 reference genome (Schneider et al., 2017) and intersected against an H1299 RNA-seq expression profile (dataset, genome indexing, expression annotation, and benchmark protocol detailed in Supplementary Methods S2.3–S2.6), siOFF completes the intersection, accessibility annotation, probability computation, and output stages 18.1−21.8× faster than the legacy Python 2.7 implementation across batch sizes of 500–2,000 siRNAs (five timed runs per batch size; Supplementary Figure S6). The speed-up comes at the cost of a larger total memory footprint: summed over all 32 workers, siOFF peaks at ∼40–45 GB versus ∼12–16 GB for the legacy pipeline (Supplementary Figure S6; worker model and memory behaviour detailed in Supplementary Methods S2.2).

Separately, the one-time genome-wide accessibility pre-computation—folding the entire hg38 reference at W=80, L=40, u=30 with 64 workers—runs ∼23% faster than the legacy `RNAplfold` script (9.0 h vs. 11.8 h) and writes profiles of the same size (192.6 GB), stored in a layout that the off-target stage memory-maps directly, at the cost of a larger memory footprint (peak proportional set size summed over all processes ∼121 GB vs. ∼18 GB; Supplementary Figure S7 and Supplementary Table S1; method detailed in Supplementary Methods S2.7).

Final Remarks

Here, we have exploited the emerging possibilities of new software ecosystems to reimplement RIsearch and the siOFF pipeline, with enhancements in installation, maintenance, integration with other tools and speed. Both tools can be used from the command line and within Python workflows by anyone screening RNA interactions or siRNA off-targets at transcriptome scale. Future work will extend siOFF to the combined off-targeting potential of pooled siRNAs and integrate RIsearch with further analysis frameworks.

Author contributions statement

S.R. reimplemented RIsearch2 in Rust and provided the framework for the siOFF pipeline. L.F. designed and implemented the siOFF pipeline, ran all benchmarks, and drafted the manuscript. C.A. designed the temperature option and tested the software. J.G. supervised the project. All authors commented on, contributed to, and approved the final manuscript.

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Conflict of interest

None declared.

Code and data availability

The source code for RIsearch and siOFF is available on GitHub (RTH-tools/risearch, RTH-tools/siOFF), and the packages on PyPI (risearch, sioff) and Crates.io (risearch). All distribution channels are summarised at rth.dk/resources/risearch. The benchmarking code and scripts used to produce the results in this paper are deposited on Zenodo (doi:10.5281/zenodo.XXXXXXX).

References

A. V. Alekseyenko and C. J. Lee. Nested Containment List (NCList): a new algorithm for accelerating interval query of genome alignment and interval databases. *Bioinformatics*, 23 (11):1386–1393, 2007. doi: 10.1093/bioinformatics/btl647.

F. Alkan, A. Wenzel, O. Palasca, P. Kerpedjiev, A. F. Rudebeck, P. F. Stadler, I. L. Hofacker, and J. Gorodkin. RIsearch2: suffix array-based large-scale prediction of RNA–RNA interactions and siRNA off-targets. *Nucleic Acids Research*, 45(8):e60, 2017. doi: 10.1093/nar/gkw1325.

O. Gotoh. An improved algorithm for matching biological sequences. *Journal of Molecular Biology*, 162(3):705–708, 1982. doi: 10.1016/0022-2836(82)90398-9.

I. Grebnov. libsais: linear-time suffix array, LCP array and Burrows–Wheeler transform construction. https://github.com/IlyaGrebnov/libsais, 2021. Accessed 2026.

D. Huesken, J. Lange, C. Mickanin, J. Weiler, F. Asselbergs, J. Warner, B. Meloon, S. Engel, A. Rosenberg, D. Cohen, M. Labow, M. Reinhardt, F. Natt, and J. Hall. Design of a genome-wide siRNA library using an artificial neural network. *Nature Biotechnology*, 23(8):995–1001, 2005. doi: 10.1038/nbt1118.

A. Lambolez, P. Sahlén, W. Kang, X. Shu, J. Severin, R. Pracana, I. Abdelhamid, B. Dhaka, C. Vroland, V. Ranzani, B. Polimeni, M. Koido, A. Vandelli, M. Mintseva, M. Rohaly Medved, K. Yasuzawa, M. Murata, D. Delobel, W. H. Yip, H. Nishiyori-Sueki, S. Takizawa, T. Nobusada, M. Brown, V. Di Gioia, Y. Inaba, S. Kato, C. Parr, K. Kaji, T. Kawashima, T. Kouno, M. Tagami, K. Ozaki, R. Vadalà, F. Marasca, E. Cozzi, R. Krautz, C. Vaagensø, T. Yamazaki, X. Li Wang, Q. Verron, Y. Ichikawa, J.-C. Chang, M. Valentine, H. Einarsson, J. Moody, A. Hasegawa, Z. Liao, K. Tomizuka, A. Zhigulev, M. Gaviraghi, Z. Altan, S. Policicchio, Y. Yu, C. Latini, R. Piccioni, C. Medaglia, A. P. Schnepper, L. Carpen, E. Ricciardelli, M. Bonfanti, R. Bosotti, C. Callierotti, G. M. Scotti, Y.-L. Chen, V. Kumar, A. Karabacak, X. Zhong, A. Naughton, I. Farabella, P. Robbe, N. Pirastu, G. Pascarella, L. P. Tripathi, et al. The human RNA-DNA interactome is cell type-specific and dynamic. *bioRxiv*, 2026. doi: 10.64898/2026.08.25.746868.

R. Lorenz, S. H. Bernhart, C. Höner zu Siederdissen, H. Tafer, C. Flamm, P. F. Stadler, and I. L. Hofacker. ViennaRNA package 2.0. *Algorithms for Molecular Biology*, 6(1):26, 2011. doi: 10.1186/1748-7188-6-26.

Y. Mori. libdivsufsort: a lightweight suffix-array construction library. https://github.com/y-256/libdivsufsort, 2015. Accessed 2026.

A. R. Quinlan and I. M. Hall. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, 26(6):841–842, 2010. doi: 10.1093/bioinformatics/btq033.

V. A. Schneider, T. Graves-Lindsay, K. Howe, N. Bouk, H.-C. Chen, P. A. Kitts, T. D. Murphy, K. D. Pruitt, F. Thibaud-Nissen, D. Albracht, R. S. Fulton, M. Kremitzki, V. Magrini, C. Markovic, S. McGrath, K. M. Steinberg, K. Auger, W. Chow, J. Collins, G. Harden, T. Hubbard, S. Pelan, J. T. Simpson, G. Threadgold, J. Torrance, J. M. Wood, L. Clarke, S. Koren, M. Boitano, P. Peluso, H. Li, C.-S. Chin, A. M. Phillippy, R. Durbin, R. K. Wilson, P. Flicek, E. E. Eichler, and D. M. Church. Evaluation of GRCh38 and de novo haploid genome assemblies demonstrates the enduring quality of the reference assembly. *Genome Research*, 27(5):849–864, 2017. doi: 10.1101/gr.213611.116.

A. Wenzel, E. Akbaşlı, and J. Gorodkin. RIsearch: fast RNA–RNA interaction search using a simplified nearest-neighbor energy model. *Bioinformatics*, 28(21):2738–2746, 2012. doi: 10.1093/bioinformatics/bts519.